PROBLEM SOLVING

DIFFERENTIAL CALCULUS

prepared by:

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

60%

EXAM

40%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

Free-falling body

Rate of Volume

Rate of Profit

PROBLEM EXERCISE

Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, **450 m** above the ground.

a. What is the **velocity** of the ball after **5 seconds**?

b. How **fast** is the ball traveling when it hits the **ground**?

The distance fallen by a freely falling body is defined by the equation :

$$s(t) = 4.9t^2$$

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The CN Tower in Toronto was the tallest free-standing building in the world for 32 years.

PROBLEM 1

Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, **450 m** above the ground.

a. What is the **velocity** of the ball after **5 seconds**?

b. How **fast** is the ball traveling when it hits the **ground**?

The distance fallen by a freely falling body is defined by the equation :

$$s(t) = 4.9t^2$$

The rate of change of distance with respect to time

$$\frac{d}{dt}s(t) = 2(4.9)t$$

$$\frac{d}{dt}s(t) = 9.8t$$

The velocity after 5s ,

$$\left.\frac{ds(t)}{dt}\right|_{t=5} = 9.8(5)$$

$$\boxed{\left.\frac{ds(t)}{dt}\right|_{t=5} = 49\ m/s}$$

ans

PROBLEM 1

Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, **450 m** above the ground.

a. What is the **velocity** of the ball after **5 seconds**?

b. How **fast** is the ball traveling when it hits the **ground**?

The distance fallen by a freely falling body is defined by the equation :

$$s(t) = 4.9t^2$$

At $s(t) = 450\ m$

$$450 = 4.9t^2$$

$$t = \sqrt{\frac{450}{4.9}} \approx 9.58\ s$$

The velocity after 9.58s ,

$$\left.\frac{ds(t)}{dt}\right|_{t=9.58} = 9.8(9.58)$$

$$\boxed{\left.\frac{ds(t)}{dt}\right|_{t=9.58} = 93.88\ m/s}$$

ans

PROBLEM 2

A **conical pile** of soil is being formed as soil pours from a spout, with the height of the cone always being twice the radius of the base. Determine the **rate** at which the **volume** of the cone is changing with respect to its **height** when the height is **4 feet**.

The volume of the cone is defined by the equation :

$$V = \frac{1}{3}\pi r^2 h$$

Let $h = 2r$

$$V = \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h$$

$$V = \frac{1}{3}\pi\frac{h^2}{4}\ h$$

$$V = \frac{\pi}{12}h^3$$

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The volume of the cone is defined by the equation :

$$V = \frac{1}{3}\pi r^2 h$$

The rate of change of volume with respect to height

$$V = \frac{\pi}{12}h^3$$

$$\frac{dV}{dh} = \frac{\pi}{4}h^2$$

$$\left.\frac{dV}{dh}\right|_{h=4} = \frac{\pi}{4}h^2$$

$$\left.\frac{dV}{dh}\right|_{h=4} = \frac{\pi}{4}(4)^2$$

$$\boxed{\left.\frac{dV}{dh}\right|_{h=4} = 12.56 \ \ cu.ft./sec}$$

ans

PROBLEM 3

A growing e-commerce company is analyzing its financial performance over time. The company observes that its revenue $R(t)$ in dollars after some time t in months is modeled by the equation :

$$R(t) = 200t^2 + 500t + 2000$$

At the same time, the company's expenses grow as it hires more employees and scales its operation, modeled by :

$$E(t) = 300t + 1000$$

Determine the rate of change of the company's **profit** after **6 months**.

Profit is revenue less expenses

$$P(t) = R(t) - E(t)$$

$$P(t) = \left(200t^2 + 500t + 2000\right) - (300t + 1000)$$

$$P(t) = 200t^2 + 200t + 1000$$

$$\frac{dP(t)}{dt} = 400t + 200$$

$$\left.\frac{dP(t)}{dt}\right|_{t=6} = 400(6) + 200$$

$$\boxed{\left.\frac{dP(t)}{dt}\right|_{t=6} = 2600 \quad \textit{dollars per month}}$$

ans

SEATWORK